Evaluation

I. Choose the best answer.

1. In which of the following Compounds does the central atom obey the octet rule?

   a) XeF_4 b) $AlCl_3$

   c) SF_6 d) SCl_2

2. In the molecule $O_A = C = O_B$, the formal charge on O_A, C and O_B are respectively.

   a) -1, 0, + 1 b) +1, 0,-1

   c) -2,0,+2 d) 0,0,0

3. Which of the following is electron deficient?

   a) PH_3 b) $(CH_3)_2$

   c) BH_3 d) NH_3

4. Which of the following molecule contain no л bond?

   a) SO_2 b) NO_2

   c) CO_2 d) H_2O

5. The ratio of number of sigma (σ) and pi (л) bonds in 2- butynal is

   a) 8/3 b) 5/3

   c) 8/2 d) 9/2

6. Which one of the following is the likely bond angles of sulphur tetrafluoride molecule?

   a) 120^0,80^0 b) $109^0.28$

   c) 90^0 d) 89^0,117^0

7. **Assertion:** Oxygen molecule is paramagnetic.

   Reason : It has two unpaired electron in its bonding molecular orbital

   a) both assertion and reason are true and reason is the correct explanation of assertion

   b) both assertion and reason are true but reason is not the correct explanation of assertion

   c) assertion is true but reason is false

   d) Both assertion and reason are false

8. According to Valence bond theory, a bond between two atoms is formed when

   a) fully filled atomic orbitals overlap

   b) half filled atomic orbitals overlap

   c) non- bonding atomic orbitals overlap

   d) empty atomic orbitals overlap

9. In ClF_3 ,NF_3 and BF_3 molecules the chlorine, nitrogen and boron atoms are

   a) sp^3 hybridised

   b) sp^3 ,sp^3 and sp^2 respectively

   c) sp^2 hybridised

   d) sp^3d, sp^3 and sp^2 hybridised respectively

10. When one s and three p orbitals hybridise,

    a) four equivalent orbitals at 90^0 to each other will be formed

    b) four equvivalent orbitals at 109^0 28' to each other will be formed.

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